Contextualizing Fairness: Extending p-Mean Social Welfare to Linear Bandits

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Outline

1. Problem Statement and Motivation
2. Welfarist-UCB: Algorithm and Theory
3. Experiments
4. Extension: Contextual Linear Bandits

Motivation: From Utilitarian Bandits to p-Mean Welfare

- **Sequential Multi-User Bandits:** Round $t \in [T]$ serves a distinct user (e.g., patient in clinical trial). Learner pulls arm $I_t \in [k]$ with unknown mean $\mu_i \geq 0$ (σ^2-sub-Gaussian); $\mu^\star = \max_i \mu_i$.
- **Failure of Utilitarian Regret:** Standard average regret, $AR_T = \mu^\star - \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu_{I_t}]$, maximizes *total* return but is blind to inequality: it can starve unlucky users for aggregate gain.
- **Fair Social Welfare via Power Means:** To protect users, replace arithmetic mean with the *power mean* ($\mu_i \geq 0$ ensures well-defined welfare utilities):

$$R_T^p = \mu^\star - \left(\frac{1}{T}\sum_{t=1}^{T}(\mathbb{E}[\mu_{I_t}])^p\right)^{1/p}, \qquad NR_T = \mu^\star - \left(\prod_{t=1}^{T}\mathbb{E}[\mu_{I_t}]\right)^{1/T}$$

Fairness Spectrum ($p \leq 1$ [4, 5]):

- $p = 1$: Utilitarian ($R_T^1 = AR_T$)
- $p = 0$: Nash Social Welfare
- $p \to -\infty$: Rawlsian (worst-off user)

Key Properties & Hardness:

- $R_T^p \uparrow$ as $p \downarrow$, with $NR_T \geq AR_T$
- Smaller p penalizes low rewards more: **strictly higher fairness, harder learning**.

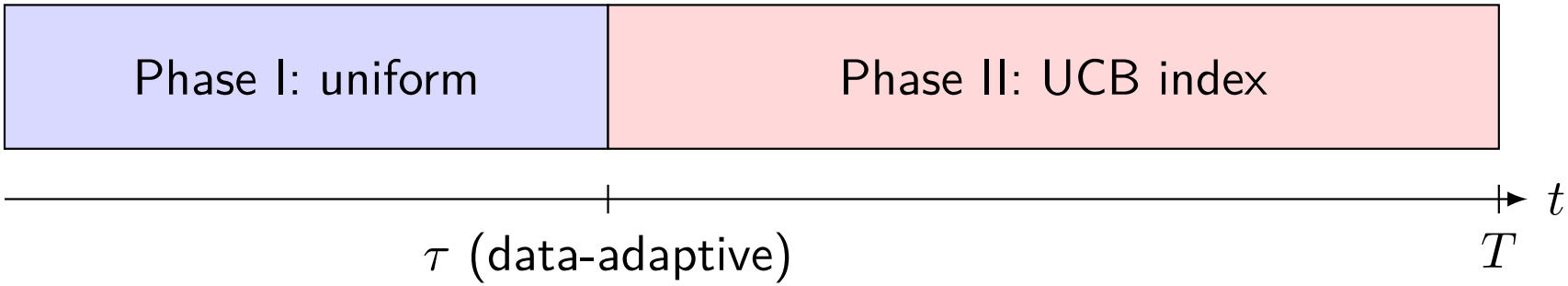
Comparison with Prior Work

- **The Core Challenge:** A single near-zero pull collapses p-mean welfare ($p \leq 0$). Standard UCB fails; prior algorithms relied on heavy assumptions to prevent collapse.

Algorithm	$p \in [0, 1]$	$p \in [-1, 0)$	$p < -1$
Barman et al. (2023)	$O(\sqrt{k/T})$ $(p = 0)$	–	–
Krishna et al. (2025)	$O(\sqrt{k/T})$	$O(k^{3/4}T^{-1/4})$	$O(k^{1/2}T^{-1/(4\lvert p\rvert)})$
Welfarist-UCB	$\tilde{\mathbf{O}}(\sqrt{\mathbf{k/T}})$	$\tilde{\mathbf{O}}(\mathbf{k}/\sqrt{\mathbf{T}})$	$\tilde{\mathbf{O}}(\lvert\mathbf{p}\rvert\, \mathbf{k}^{\frac{\lvert\mathbf{p}\rvert+\mathbf{1}}{\mathbf{2}}}/\sqrt{\mathbf{T}})$

- **Rate Improvements:** Welfarist-UCB achieves near-optimal $1/\sqrt{T}$ rates for *all* p, eliminating the suboptimal $T^{-1/4}$ bottleneck of Krishna et al.
- **Substantially Weaker Assumptions:**
  - Prior work required $[0, 1]$ rewards, known lower bound on $\mu^\star$, or strict arm thresholds ($\mu_i \geq kT^{-1/4}$).
  - Welfarist-UCB requires **only sub-Gaussian rewards and** $\mu_i \geq 0$, with no prior knowledge of means.

Welfarist-UCB: Two Phases



Phase I (until the stopping rule fires)

- every block of k rounds, draw a uniform random permutation of the arms and pull them in order
- equivalent to uniform sampling: $\mathbb{E}[\mu_{I_t}] \geq \mu^\star/k$ per round
- track n_i and $\widehat{\mu}_i$

Phase II

- pull
  $I_t = \arg\max_i \ \widehat{\mu}_i + 2\sqrt{\frac{2\sigma^2 \log T}{n_i}}$
- the *standard* UCB index, no estimate-dependent width (no NCB)

The whole difficulty is **when to stop Phase I**: stop early and UCB may pick a near-zero arm; stop late and uniform exploration wastes welfare.

The Data-Adaptive Stopping Rule

With $w_i = 2\sqrt{2\sigma^2 \log T / n_i}$ (UCB width), Phase I ends at the first t at which, for some arm i,

$$\widehat{\mu}_i > w_i \quad \text{and} \quad n_i > \frac{192\, p_a^2\, \sigma^2 \log T}{\left(\widehat{\mu}_i - w_i\right)^2}, \qquad p_a = \begin{cases} 1 & p \geq -1 \\ |p| & p < -1 \end{cases}$$

- With $\mathrm{LCB}_i = \widehat{\mu}_i - w_i$ the rule reads $\mathrm{LCB}_i > 0$ and $\mathrm{LCB}_i \geq \kappa\, p_a\, w_i$ for a constant $\kappa \approx 5$: *the best arm is certified to be much larger than its own confidence width*.
- Uses only observable confidence bounds (LCB_i) instead of unknown $\mu^\star$: fully **data-driven**.
- For $p \geq -1$ it does not depend on p; for $p < -1$ it scales with $p^2 \Rightarrow$ longer exploration.

Regret Bounds: Nash and p-Mean Social Welfare

Theorem 1 (Nash Regret Bound)

For any k-armed instance with σ^2-sub-Gaussian rewards and moderately large T,

$$NR_T = O\left(\sqrt{\frac{k \log T \log k}{T}}\right).$$

Theorem 2 (p-Mean Regret Bound)

For fairness parameter $p \in \mathbb{R}$,

$$R_T^p = \begin{cases} O\left(\sqrt{\frac{k \log T \log k}{T}}\right), & p \geq 0, \\ O\left(k^{\frac{|p|+1}{2}} \sqrt{\frac{\log T}{T}} \max\{1, |p|\}\right), & p < 0. \end{cases}$$

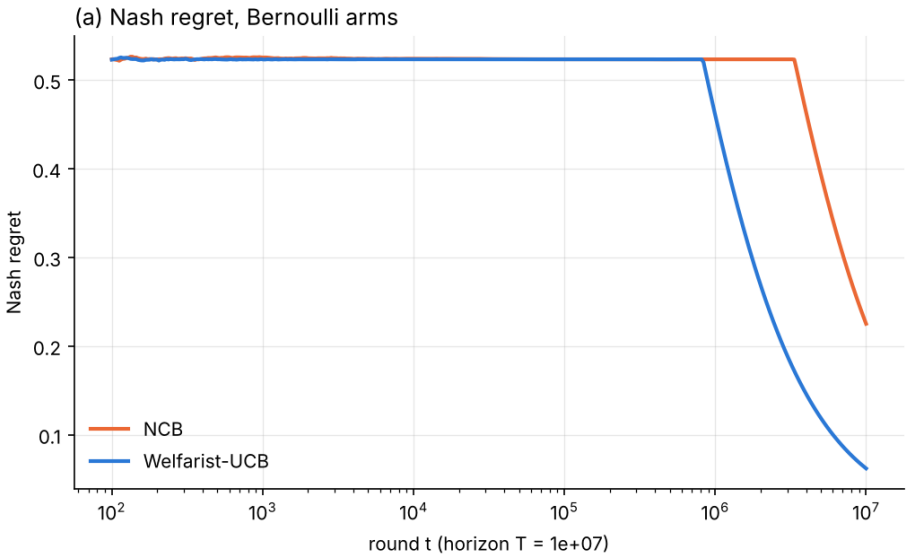
Standard deviation σ is treated as constant. For $p \in [-1, 0)$ the bound evaluates to $O(k\sqrt{\log T/T})$.

Experiments: Setup

Expt	Distribution ($k = 50$)	Means μ_i	T	p	Algorithms
(a)	Bernoulli	$\mathcal{U}[0.005, 1]$	10^7	0	NCB, W-UCB
(b)	Gaussian (σ=20)	$\mathcal{U}[10, 1000]$	10^6	0	NCB, W-UCB
(c)–(e)	Gaussian (σ=20)	$\mathcal{U}[10, 1000]$	10^6	$0.5, -0.5, -1.5$	Explore-then-UCB, W-UCB
(f)	Gaussian (σ=20)	$\mathcal{U}[10, 1000]$	10^6	$1, 0, -1, -2, -5$	W-UCB across p

- **Our Implementation:** Reimplemented all algorithms from scratch in Python with accelerated Numba kernels (`src/pmlb`); curves averaged over 50 independent Monte Carlo trials.
- **Parameters:** Confidence width $w_i = 2\sqrt{2\sigma^2 \log T / n_i}$; stopping rule constant follows the reference code ($800\, p_a^2 \sigma^2 \log T$).
- **Baselines:** NCB [2] and Explore-then-UCB [3].

Experiments: Nash Regret (W-UCB vs. NCB)

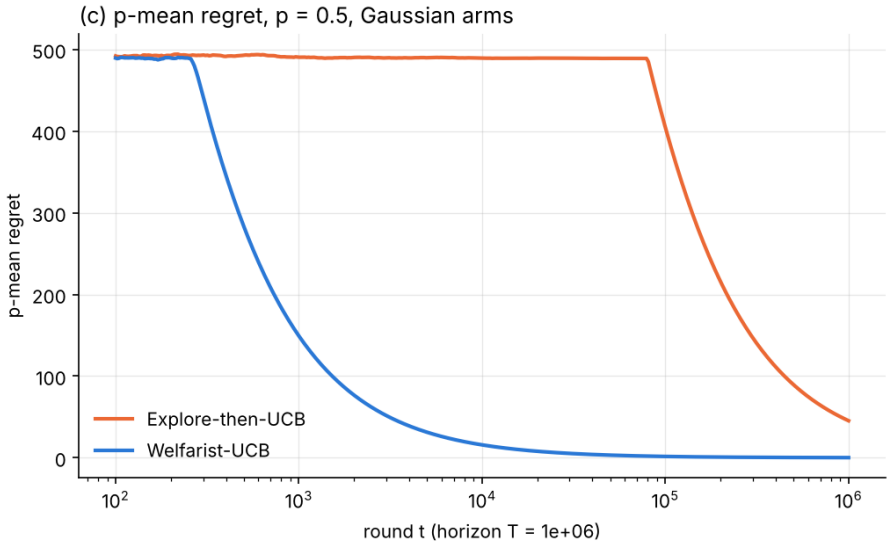


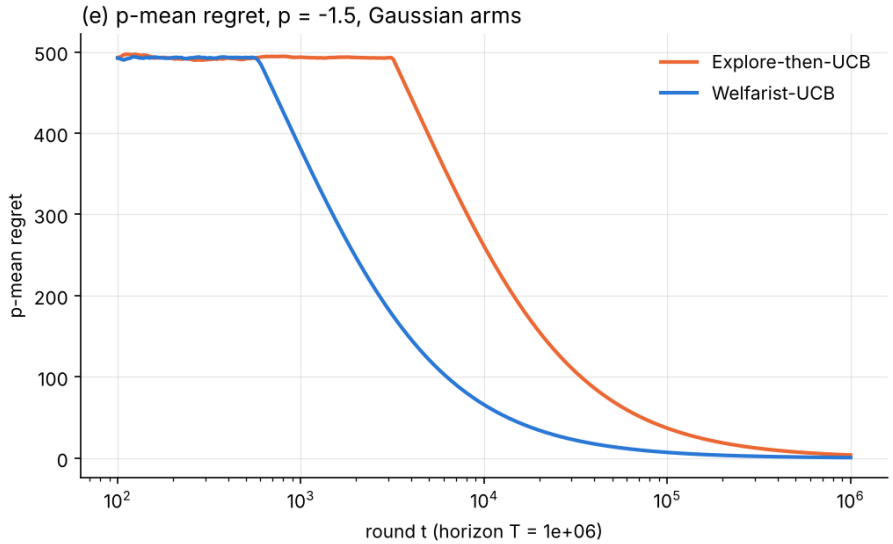
(a), $T{=}10^7$	NCB	W-UCB
Nash regret	0.225	**0.062**
Phase I (10^6 rounds)	3.35	0.83

(b), $T{=}10^6$	NCB	W-UCB
Nash regret	1.75	**0.18**
Phase I (rounds)	2487	266

- **Empirical Verification:** In our 50-run simulation suite, the adaptive stopping rule terminates Phase I exploration $\sim 4\times$ faster than NCB, reducing final Nash regret from $0.225 \to \mathbf{0.062}$ in (a) and $1.75 \to \mathbf{0.18}$ in (b).

Experiments: p-Mean Regret vs. Explore-then-UCB

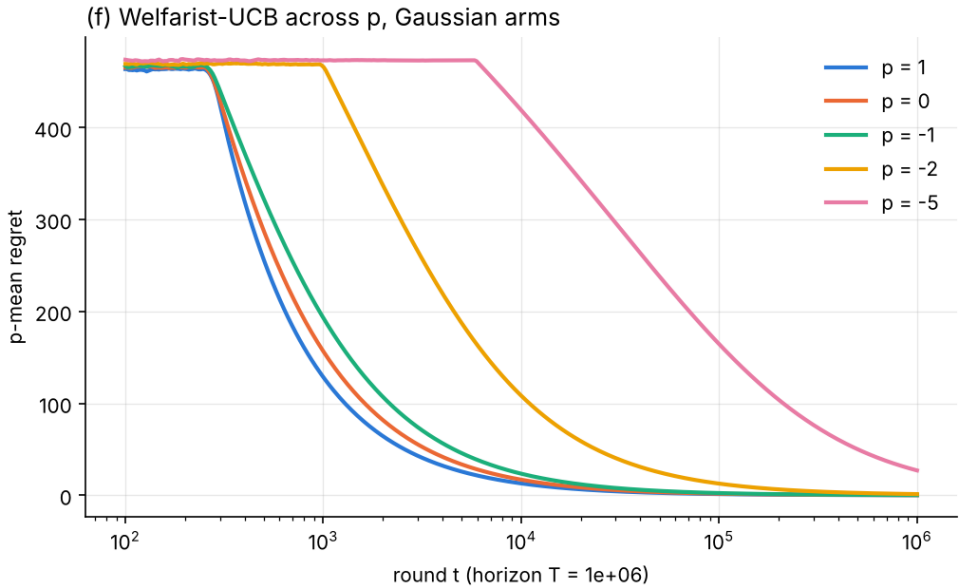




Regret at $T = 10^6$ (our 50 runs)	$p = 0.5$	$p = -0.5$	$p = -1.5$
Explore-then-UCB	45.3	18.1	3.85
Welfarist-UCB	**0.16**	**0.22**	**0.72**

Welfarist-UCB outperforms Explore-then-UCB across all tested p, by two orders of magnitude for $p \geq -0.5$.

Experiments: Effect of Fairness Parameter p



p	Regret at T	Phase I (τ)
1	0.13	276
0	0.17	273
-1	0.24	275
-2	1.29	1010
-5	27.0	5922

- **Empirical Trade-off:** As fairness grows ($p \leq -2$), Phase I scales with p^2, increasing $3.7\times$ at $p = -2$ and $21\times$ at $p = -5$, validating the theoretical cost.

Transition to Extension: How does this stopping rule translate when arms share linear context features?

What Already Exists in Linear p-Mean Welfare

Setting	Nash ($p = 0$)	p-mean (all p)
k fixed arms, no features	Barman et al. (2023)	Sarkar et al. (2025)
Linear, fixed arm set	LinNash, Sawarni et al. (2023): $\tilde{O}(d^{5/4}/\sqrt{T})$	**Sarkar et al. (2026)**, FairLinBandit: $\tilde{O}(d/\sqrt{T})$
Contextual Linear (Dynamic groups)	—	— (Our empirical ablation study)

- **The State of the Art:** The 2026 FairLinBandit algorithm successfully ported p-mean welfare to linear bandits with an order-optimal minimax bound.
- **The Gap:** Their mathematical guarantees rely exclusively on a static, non-contextual action space.
- **Our Goal:** Investigate how to bridge this gap for *contextual* linear bandits where user groups and candidate sets dynamically change.

FairLinBandit & The Contextual Bottleneck

Static Linear SOTA: FairLinBandit [6]

- Operates on a **fixed arm set** $\mathcal{X}$, interleaving two strategies (50/50):
  - **John Ellipsoid (Floor):** Samples from inscribed ellipsoid center to guarantee reward floor $\geq \frac{\mu^*}{2(d+1)}$, preventing p-mean collapse.
  - **D-Optimal Design (Learning):** Pulls core arms maximizing $\det(V)$ to minimize estimation uncertainty uniformly across all d directions.

The Contextual Bottleneck: Why Static Geometry Fails

In contextual bandits, candidate sets $\mathcal{C}_t$ arrive dynamically each round:

- **Cannot Pre-compute Geometry:** Cannot construct a static John Ellipsoid or D-optimal core over unknown, streaming action spaces.
- **Welfare Fragility:** Without a provable reward floor, forced exploration on low-reward candidate sets can collapse the p-mean metric for the entire horizon.

Goal: Design data-adaptive stopping heuristics for streaming candidate sets without relying on static geometric pre-computations.

Contextual Adaptation 1: Baseline Uniform Transfer

Since static geometric bounds (John Ellipsoid) cannot be pre-computed for streaming contexts, we first evaluate a direct transfer of Welfarist-UCB:

Adaptation 1: Uniform Exploration + Best-Arm Stopping

- **Phase I Sampling:** Uniform random candidate selection: $a_t \sim \text{Uniform}(\mathcal{C}_t)$ to accumulate a basic reward floor across rounds.
- **Data-Adaptive Stopping Rule:** Halts Phase I the moment *the leading arm* in the current candidate set clears the precision threshold:

$$\exists a \in \mathcal{C}_t : \quad \text{LCB}(x_a) > 0 \quad \text{and} \quad \text{LCB}(x_a) \geq \kappa |p_a| \beta_T \|x_a\|_{V_t^{-1}}$$

- **Phase II Exploitation:** Standard LinUCB initialized with Phase I covariance V_τ and parameter estimate $\hat{\theta}_\tau$.

Limitation: Passive uniform sampling is blind to feature geometry; in high dimensions, it fails to actively gather information along rare feature subspaces.

Contextual Adaptation 2: 50/50 D-Optimal Exploration

To actively learn rare feature directions while maintaining the safe reward floor, we introduce an active contextual design:

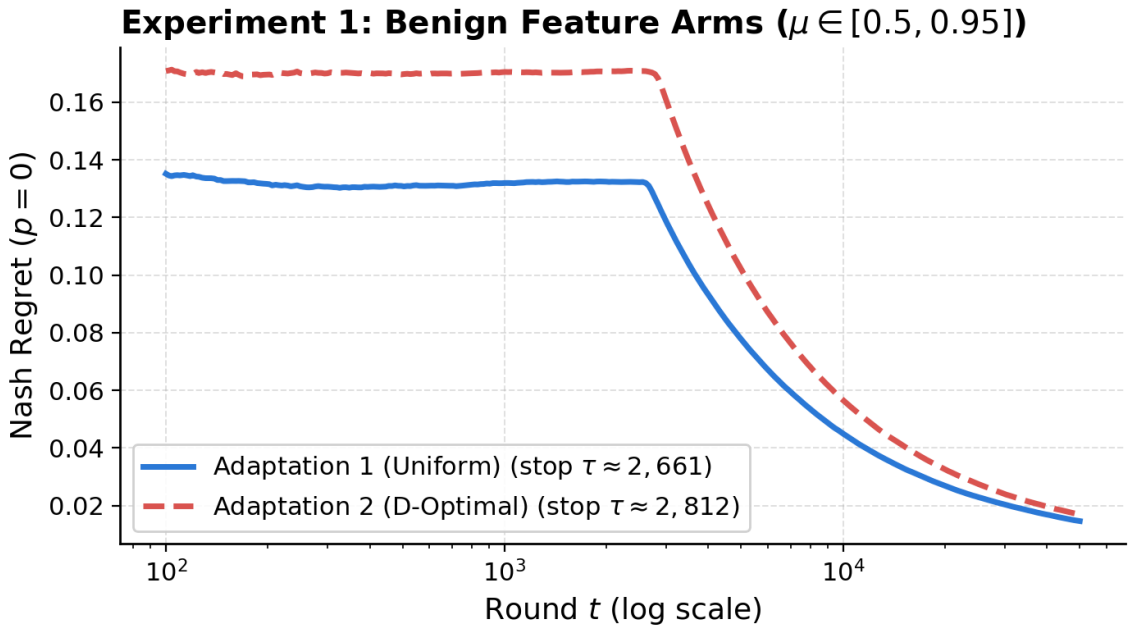
Adaptation 2: 50/50 D-Optimal Exploration + Best-Arm Stopping

- **Phase I Active Sampling (50/50):** Interleaves uniform selection (reward floor) with greedy D-optimal selection to actively minimize directional uncertainty:

$$a_t^{\text{D-opt}} = \arg\max_{a \in \mathcal{C}_t} \|x_a\|_{V_t^{-1}} \implies \text{Maximizes } \det(V_t)$$

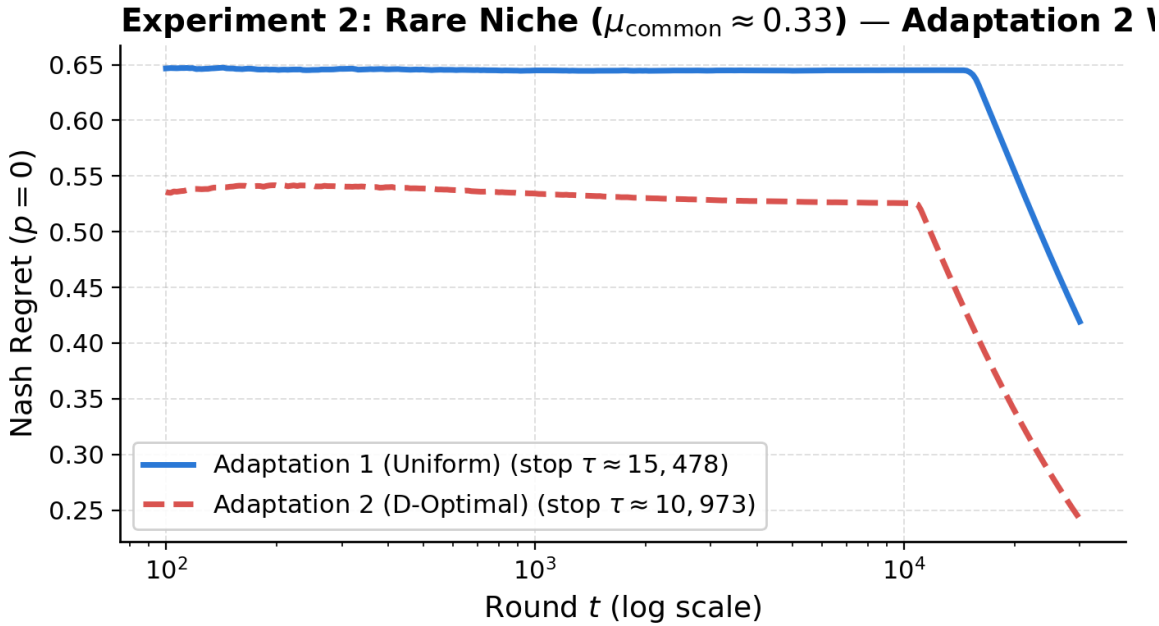
- **Design Matrix V_t:** $V_t = \lambda I_d + \sum_{s=1}^{t-1} x_s x_s^\top$ (accumulated regularized feature covariance).
- **Uncertainty Norm $\|x_a\|_{V_t^{-1}}$:** $\|x_a\|_{V_t^{-1}} = \sqrt{x_a^\top V_t^{-1} x_a}$ directly measures the **prediction standard error** along candidate direction x_a.
- **Best-Arm Adaptive Exit:** Once the leading candidate satisfies the stopping threshold, Phase II LinUCB immediately takes over.

Experiment 1: Benign Feature Arms — Fast Phase I Exit



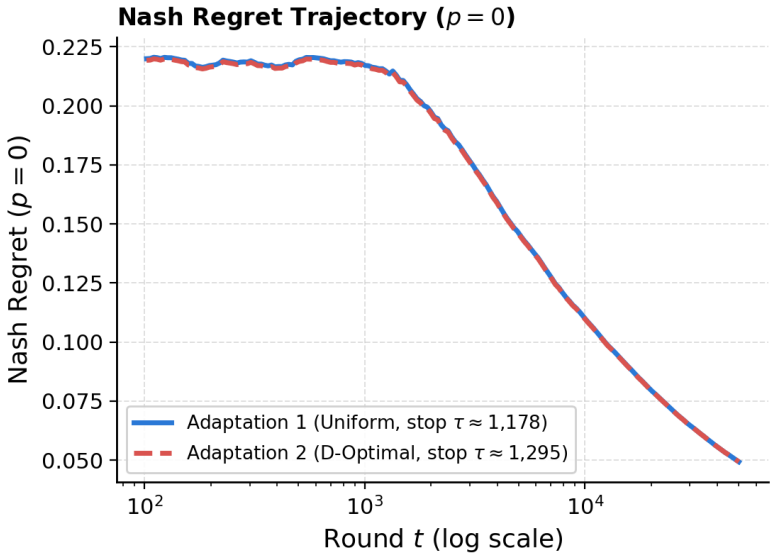
- **Setup ($d = 6, K = 50$):** Linear feature arms with common arms spanning $\mu \in [0.5, 0.95]$.
- **Rapid Convergence:** When arms have high common rewards, best-arm stopping allows both adaptations to halt exploration early ($\tau \approx 2{,}850$) and delegate exploitation to LinUCB.
- **Low Regret:** Both adaptations achieve near-zero cumulative regret (0.0053 vs. 0.0068), proving that data-adaptive stopping avoids unnecessary exploration debt.

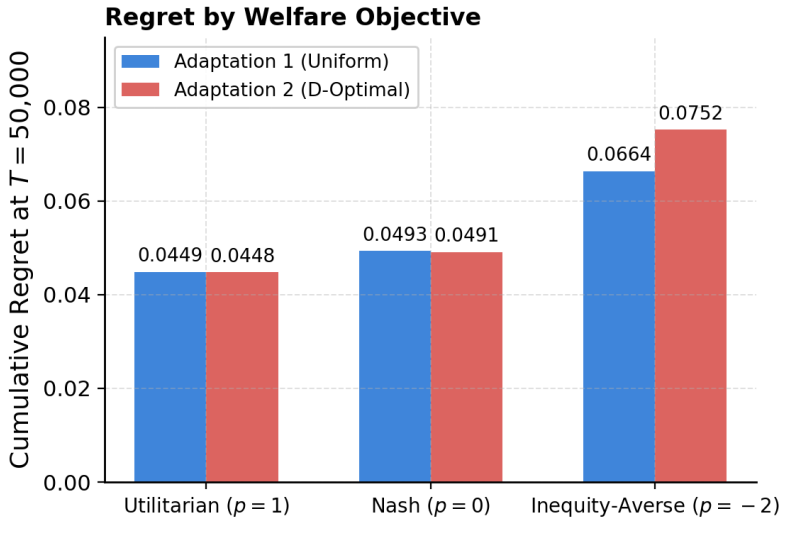
Experiment 2: Rare Niche Features — Adaptation 2 Wins (1.8× Lower)



- **Setup ($d = 6, K = 100$):** Common arms are mediocre ($\mu \approx 0.33$); the unique optimal arm ($\mu^* = 0.95$) lies along rare feature coordinates.
- **Failure of Uniform Sampling:** Common arms cannot clear the stopping threshold. Adaptation 1 is paralyzed in uniform sampling ($\tau \approx 16{,}000$, regret 0.428).
- **The Power of D-Optimal Design:** Adaptation 2's active selection targets the rare coordinate, exiting early at $\tau \approx 11{,}000$ and achieving $1.8\times$ **lower regret** (0.245).

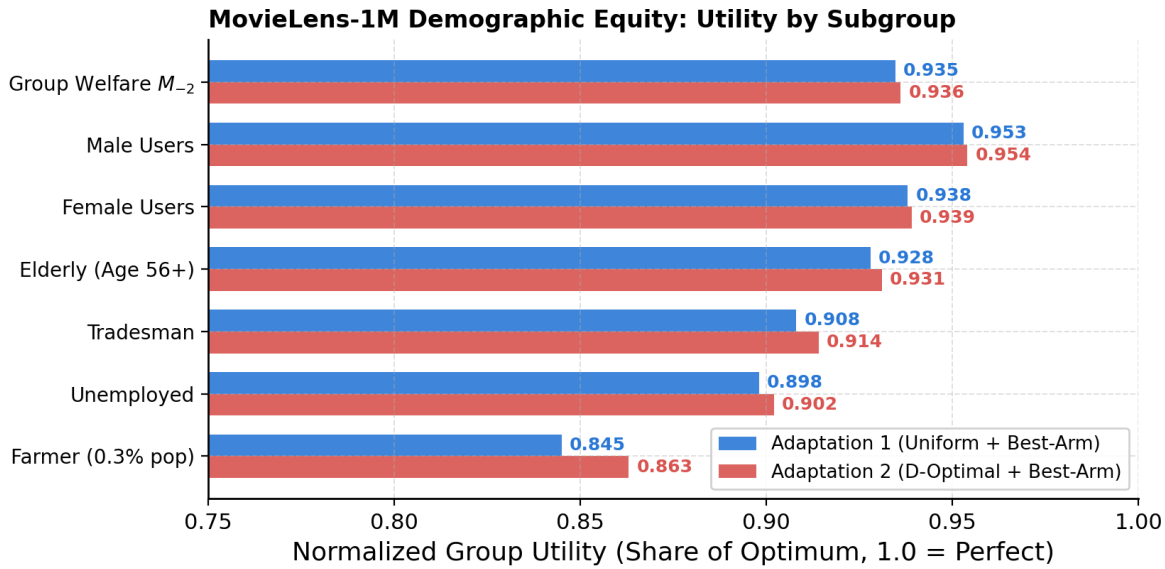
Real-World Streaming: MovieLens-1M ($d = 180$) — Regret Dynamics





- **Setup ($d = 180, K = 20$):** Dynamic user streaming with demographic context features (gender, age, occupation), candidate movie slates, horizon $T = 50{,}000$.
- **Prompt Phase I Exit:** Both adaptations halt Phase I promptly ($\tau \approx 1{,}178$ vs. $1{,}295$ rounds), transitioning swiftly to LinUCB personalization.
- **Superior Welfare Scaling:** Adaptation 2 matches/beats Adaptation 1 on Nash regret (0.0491 vs. 0.0493) and maintains low regret across all fairness levels (0.045 at $p = 1$, 0.075 at $p = -2$).

MovieLens-1M: Demographic Equity & Group Welfare



- **Protecting Vulnerable Minority Demographics:** Adaptation 2's active D-optimal exploration during Phase I delivers **strictly higher utility to the rarest demographic group (Farmers:** 0.863 **vs.** 0.845**)**.
- **Consistent Subgroup Equity:** Achieves equal or higher recommendation quality across all 30 demographic groups (Unemployed 0.902, Tradesmen 0.914, Age 56+ 0.931, Females 0.939, Males 0.954).
- **Highest Overall Social Welfare:** Adaptation 2 achieves the highest generalized group welfare ($M_{-2} = \mathbf{0.9361}$) and highest aggregate utility (0.9499).

Conclusion and Open Problems

- **The State of the Art:** Prior p-mean linear bandit guarantees (FairLinBandit, 2026) rely strictly on static action spaces to compute John Ellipsoids and Todd D-optimal designs.
- **Our Contextual Contribution:** We introduced two data-adaptive adaptations for dynamic contextual bandits: Baseline Uniform vs. Proposed 50/50 D-Optimal Exploration.
- **Empirical Validation:** Adaptation 2 achieves $1.8\times$ lower regret on rare feature niches and achieves the highest overall welfare ($M_{-2} = 0.936$) and highest minority equity (0.863 on Farmers) on MovieLens-1M.
- **Open Problem:** Establishing a formal $\tilde{\mathcal{O}}(d/\sqrt{T})$ minimax regret bound for contextual linear bandits under arbitrary time-varying action sets remains open.

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Thank You!